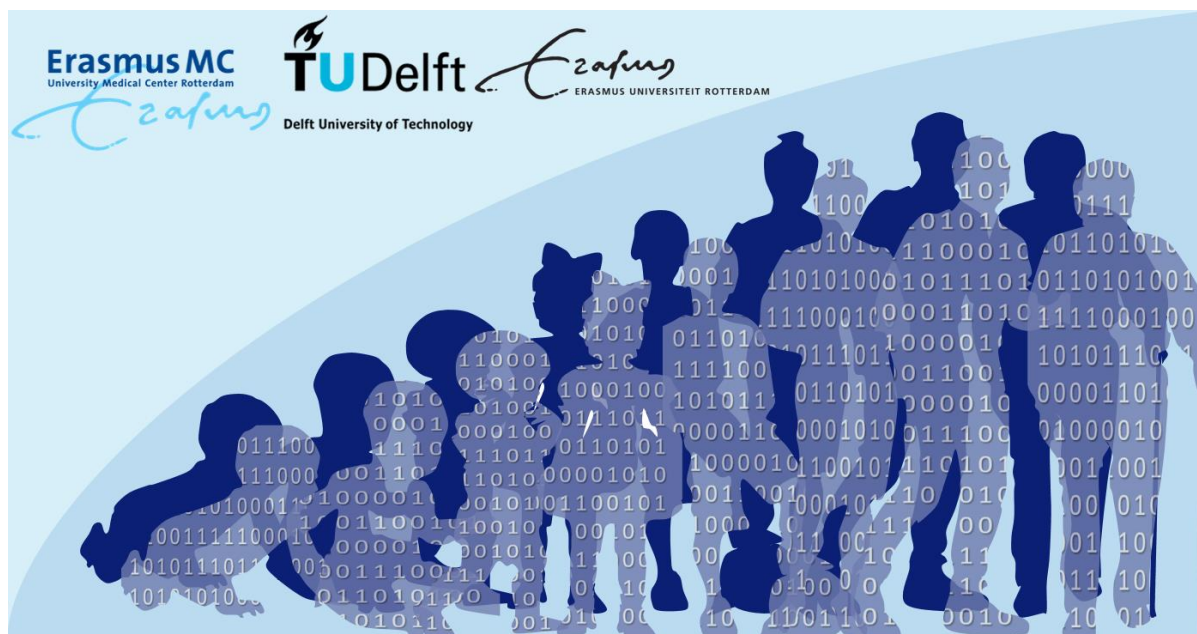




"HEALTHY AGEING ACROSS THE LIFE COURSE": FROM BENCH TO BEDSIDE AND INTO SOCIETY (HELIOS)

1. General information

1.a Applicants

¹ Convergence Digital Twin project

Name*	Institution	Department	Email	Tenured (i.e. fixed) position/ Tenure-track/ Other	For Erasmus MC applicants: clinician/ researcher/ mix	Role in Flagship [Main Lead/Lead/Applicant]
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WP3						
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C. Sportelli	Avegen	Avegen Health	carolina@avegenhealth.com	Private Partner	Company
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Planned start date of seed funding: 1-11-2022

Primary Convergence theme: *Technology Supported Transitions in Health(care)*

Secondary Convergence theme (if applicable): *Improving Health Journeys*

Tertiary Convergence theme (if applicable): *Human Centered Technology and AI for Health*

Quaternary Convergence theme (if applicable): *Fundamentals of Health & Disease*

Keywords: Healthy ageing, prevention, lifestyle, health inequity, implementation, artificial intelligence, ecosystem/environment, cardiovascular disease, women's health

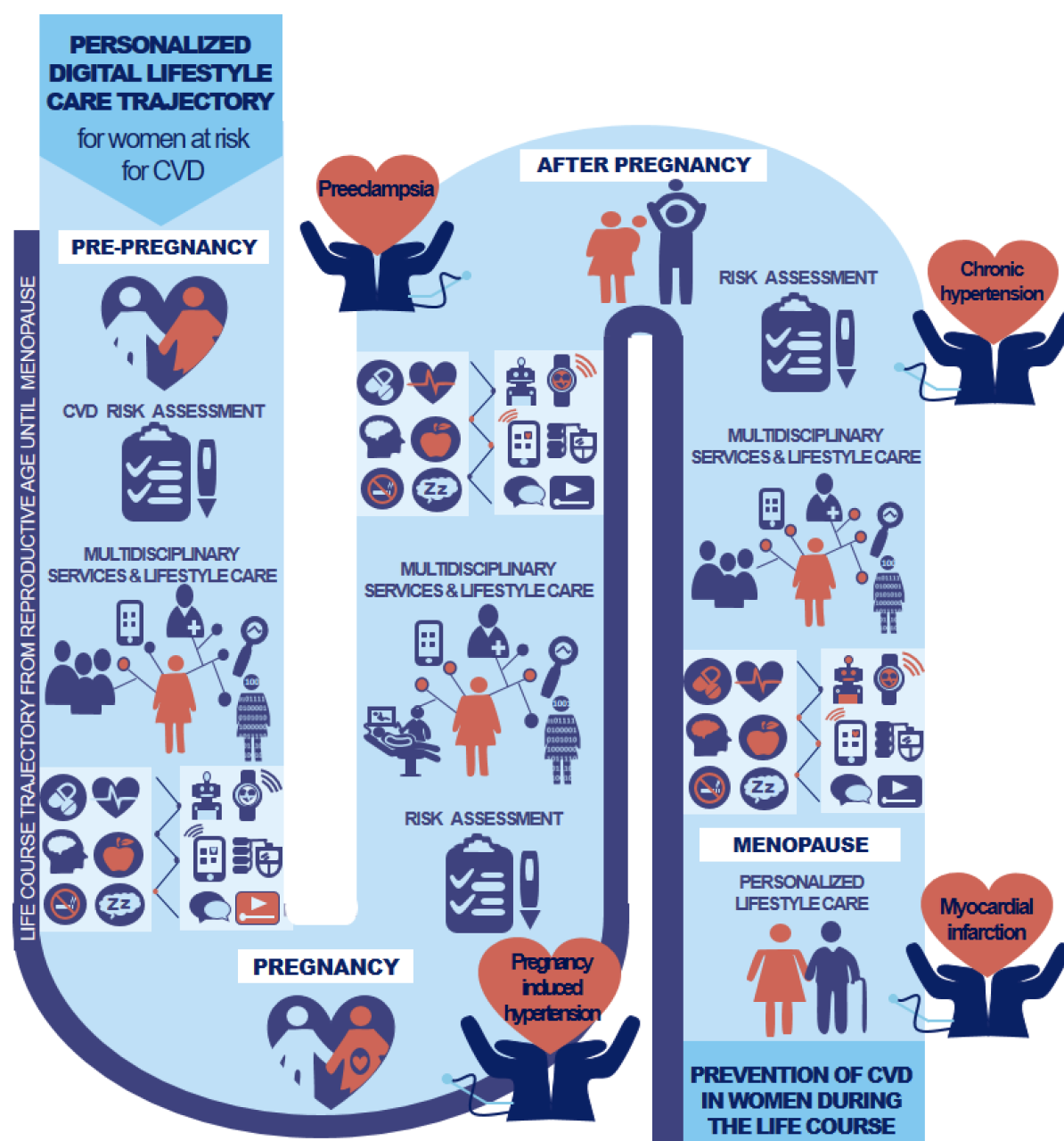


Figure1.

Personalized digital Lifestyle Care Trajectories (LCT) for prevention of CVD in women across the life course

1.b Summary

The overarching aim of the '*Healthy Ageing Across the Life Course*' flagship (HELIOS) is to **generate significant health improvements in lifestyle-related-ageing diseases through an integrative, informative and societal lifecourse care** approach. HELIOS will transform healthcare by a holistic sustainable approach including lifestyle and social care across the life course using state-of-the-art research and technology.

Our consortium of quadruple helix partners will establish a **complex, multi-modal process**, whereby multiple fixed (age, gender, genome, ethnicity), modifiable (lifestyle, including nutrition), medical conditions (physical, mental) and environmental risks (accidents, pollution, infection: Covid-19) are gathered. Using sophisticated artificial intelligence (AI), we will analyze Big data and novel knowledge for further development of the ErasmusMC '**Digital Connected**' platform. Together with co-design strategies, we will use this platform to develop effective '*healthy ageing*' apps, wearables and tele-monitoring devices that define *personalized digital Lifestyle Care Trajectories* (LCT) as roadmap for medical and social care, together serving as 'life course care journeys'. To ensure implementation and compliance, interactive consultation methods, Avatars, and loyalty programs will be developed and blended into these LCT to empower citizens to stay healthy across the life course. This will lead to fewer healthcare visits and hospital admissions, lower healthcare and societal costs, and improved wellbeing.

The first project will focus on the '*prevention of cardiovascular diseases (CVD) in women across the life course*', investigating lifestyle, healthcare, family and (social) environment using AI and co-creation techniques (Figure1). LCT will be developed for digital screening and prediction of personal CVD risks followed by offering preventive personalized lifestyle care. Success will be measured by cardiovascular (American Heart Association's CardiovascularHealthScore), value-based healthcare (ICHOM), and implementation outcomes. This novel Health&Tech innovation will be applied for the **long-term strategy of LCT development and implementation in healthcare and society for prevention of lifestyle-related-ageing diseases affecting all ages and both genders.**

1.c Public Summary

Het doel van het onderzoeksprogramma "Gezond ouder worden gedurende de levensloop" is het bereiken van significante gezondheidsverbeteringen voor leefstijl-gerelateerde ouderdomsziekten door het ontwikkelen en aanbieden van integratieve, informatieve en maatschappelijke zorg in de vorm van 'levensloopzorg'. Met een holistische, duurzame aanpak, inclusief leefstijl en sociale zorg, transformeren wij de gezondheidszorg. Hiervoor worden state-of-the-art onderzoek en nieuwe en bestaande technologieën gebruikt.

Ons consortium bestaat uit onderzoekers, burgers, overheid en bedrijven, die in een complex, multimodaal proces vaste (leeftijd, geslacht, genen, etniciteit), beïnvloedbare (leefstijl, inclusief voeding) en medische condities (lichamelijk, mentaal) en omgevingsrisico's (ongelukken, vervuiling, infecties: Covid-19) verzamelen en meten. Met geavanceerde kunstmatige intelligentie zullen we de Big data analyseren en de nieuwe kennis gebruiken voor het verder ontwikkelen van het ErasmusMC 'Digitaal Verbonden' platform. In co-creatie zullen we dit platform gebruiken om effectieve 'gezond ouder worden' apps, wearables en tele-monitoring apparaten te ontwikkelen en implementeren in persoonlijke digitale Leefstijl ZorgPaden (LZP), die worden gecombineerd met medische en sociale zorgpaden. Om het langdurig gebruik te bevorderen, zullen interactieve consultatiemethoden, Avatars en spaarprogramma's worden ontwikkeld en geïmplementeerd en via LZP worden aangeboden. Hiermee zullen burgers langer gezond blijven en zal de gezondheidszorglast verminderen (minder consulten en opnames, lagere zorgkosten).

Het eerste project (Figuur1) is gericht op de "preventie van hart- en vaatziekten bij vrouwen gedurende de levensloop", waarbij nieuwe kennis wordt gegenereerd met kunstmatige intelligentie

uit de verzamelde gegevens over leefstijl, gezondheid, ziekten, familie en sociale omgeving. In co-creatie worden LZP ingericht voor het digitaal screenen en voorspellen van hart- en vaatziekten risico's, gevolgd door persoonlijke preventieve leefstijlzorg. Succes zal worden gemeten voor cardiovasculaire, waarde-gedreven zorg, en implementatie uitkomsten. Deze nieuwe kennis over gezondheid en techniek zal worden gebruikt voor de lange termijn ontwikkeling van LZP voor preventie van leefstijl-gerelateerde ouderdomsziekten op alle leeftijden en voor mannen en vrouwen.

2. Relevance to Convergence Health & Technology mission

(50%) – max. 2500 words for 2.a to 2.e (total word count 2416)

2.a Contribution to H&T mission and fit with H&T theme(s)

The HELIOS program will contribute to all 4 H&T convergence missions:

1. *Life-long health.* The 5 years' aim is to develop personalized digital Lifestyle care trajectories (LCT) for women at risk of cardiovascular diseases (CVD), first targeting the critical periods around pregnancy and menopause to prevent CVD later in life, but also in offspring and even in next generations. This is in accordance with the Developmental Origins of Health and Disease paradigm (DOHaD). After 5 years, it is our ambition to extend LCT to other lifestyle-related ageing-diseases, e.g., diabetes, cancer and dementia, across the life course and for both genders.
2. *Socio-economic equality.* Health, lifestyle, and socio economic circumstances are strongly correlated with, and mediators of a wide range of age-related-diseases, including CVD. Moreover, these determinants are essential in the pathway of health inequality and inequity already starting around pregnancy for women and offspring with continuation across the life course and even transgenerational. Hence, reducing the accumulation of lifestyle risk factors in especially vulnerable women will be the best investment with impact on health, quality of life (QoL), healthcare and society. Therefore, we will explicitly develop and implement LCT also for these women.
3. *Prevention and early diagnostics.* The program centers around gathering data, analyzing, co-design of screening and interventions, and the implementation of LCT for prevention, prediction, monitoring and early diagnosis of CVD. We target the pregnancy and menopause period for the earliest prevention and diagnosis.
4. *Individual-tailored medical treatment.* LCT will offer personalized lifestyle interventions in life course journeys in combination with medical and societal care and tailored to the needs and values of women at risk of CVD. Thus, these H&T innovations contribute to individual-tailored medical treatment.

2.b Synergy between disciplines

The quadruple helix partners of HELIOS (researchers, citizens, companies, governmental organizations) bring together all expertise and knowledge required for LCT in life course care journeys. This synergism and complementarity is illustrated in the workpackages (WPs) and provides critical factors for success of the development, testing, feasibility and implementation of LCT in healthcare:

- **WP1:** Molecular and developmental biologists, epidemiologists, nutritionists and clinicians will select key CVD and ageing biomarkers and effective interventions for CVD in women, and transfer their insights to WP2-5.

- **WP2:** Design and technical co-creation strategies with a diversity of end-users and stakeholders will enhance the synergy with HELIOS designers, clinicians, researchers, technicians, behavioral-ICT- and communication experts.
- **WP3:** AI and data science experts will work on algorithms with designers, researchers and clinicians on multi-level, multidisciplinary data to generate human-centered CVD risk prediction scores taking into account interoperability and legal issues related to the data.
- **WP4:** LCT testing and implementation in healthcare will be a joint effort of clinicians, epidemiologists, communication experts and women with CVD, their families and stakeholders.
- **WP5:** Embedded ethicists will implement a design for values approach across the WPs, specifically focusing on meaningful human control in the recommendations generated by the algorithms in WP3, and ethics of interpersonal influence and patient autonomy in WP2/WP4. The experts on health policy and management will model cost-effectiveness and address costs of WP2/WP3 and for WP4 outcomes.

The specialists in Communication Design for Innovation (CDI) from TUD, who are currently supporting us in the Convergence Digital Twin project, will continue with the steering of these types of complex collaborations.

2.c Long term sustainability plan

Feasibility first 5-years

- 1) End-users: Women at risk of CVD are motivated to use LCT, because of their benefits from effective lifestyle care for reproduction (short-term), QoL and lifetime expectancy (long-term). We will use the infrastructure of our ongoing cohorts in WP2 as living labs for co-design and testing and in WP4 for implementation of LCT in healthcare and society (Rotterdam Periconception Cohort >2009, Generation-R>2006, Generation-R-Next>2017, FollowUpPreeclampsia>2011 (FUPEC).
- 2) Data: Access to unique genome, molecular biology, medical, lifestyle and environment data, and available knowledge, skills and platform for Big data analysis in a secured environment, are in place.
- 3) Expertise: HELIOS builds on experiences and synergistic collaborations of our current Convergence Digital Twin project, including the development of co-creation strategies with end-users and stakeholders for high-tech applications, extensive technological and AI expertise, >15 years-experience of developing lifestyle care innovations (SmarterPregnancy.co.uk, blended LCT 'SlimmerGezond.nl').
- 4) Education: Together with educational experts from the Graduate school ErasmusMC, EUR, TUD, Rotterdam University Applied Sciences, municipality Rotterdam and event-organization 'Improve', we will create educational programs and learning communities for end-users, including caregivers and stakeholders involved in lifestyle care and its implementation in healthcare and society.
- 5) Techniques: We will use available and effective Health&Tech tools, devices and AI, and develop new products and interventions together with Microsoft-NL, Microsoft-Europe and other ICT companies.
- 6) Implementation: A pre-existing implementation strategy/plan is already in place, including an evaluation and dissemination loop and validation process for LCT. We will use the ErasmusMC Digital Connected platform for integrating LCT and for implementation with medical and social care journeys.
- 7) Finances: The already committed 3,45 million co-financing, the potential commitments for co-financing and mutual grant applications by multiple stakeholders confirm the financial feasibility of HELIOS in the first 5-years and as long-term investment.
- 8) Operability&ICT: The operability of different ICT-systems will be solved by using the ErasmusMC Digital Connected as platform on which available and newly developed E-apps, Avatars, and devices will be integrated. ErasmusMC Information and Technology supports HELIOS and on board are large ICT companies and start-ups, i.e., Microsoft, Avegen, Corsano, Somnox.
- 9) Implementation: We will manage ethical, health economic, societal and legal issues through an

ethical framework, business- and strategic implementation plan. LCT will be the entrance also for the Flagship applications “Continuity of care from hospital to home” and “Human-centered AI for sustainable critical care”, further enhancing implementation feasibility.

Long-term sustainability

LCT for women at risk for CVD, family and environment will be a blueprint for expanding to other lifestyle-related-ageing diseases and also to men of all ages. Moreover, as the ErasmusMC has already collaborated for >5 years with 8 other like-minded university hospitals in the European University Hospital Alliance (www.euhalliance.eu), we are able to extend LCT to a pan-European audience and aim for large-scale implementation across Europe. Finally, with the aim to open the platform, we believe that our research results will become more accessible and reusable to the broader research community compared to leveraging proprietary technology.

Challenges

Short- and long-term challenges are to keep knowledge and experts in the HELIOS consortium and to achieve a sustainable funding level. Therefore, we will continue at least 2 tenure track positions from the Convergence Digital Twin project to keep talented people within our organizations. Moreover, HELIOS will develop education programs and generate grant-proposals as important tasks for continuity. Finally, we will continue to expand our collaborations with private and public partners, and our delivered Health&Tech innovations will be exploited nationally and internationally, again leading to an important source of income.

2.d Visibility, attracting talents and new partners

Visibility

We will launch the HELIOS website and newsletter for networking and external stakeholders, i.e., patients, families, ambassadors of the board of clients, patient organizations (HELLP Foundation), Dutch society for CVD, schools, municipality of Rotterdam. Official social media accounts will be created for broad outreach. To connect more with our lay audience, we will prepare with our partner event-organization ‘Improve’ project-related educational materials, such as leaflets, posters, and short videos, that will be shared in our official social media accounts. After 5 years, we will organize a national event to present end-users and stakeholders our experiences and outcomes, and our long-term roadmap. In addition, we will present HELIOS and its results at several conferences and meetings of academic and practice communities in which the consortium members are already involved, i.e., international consortium of value-based health care, DoHAD conferences, Conference on Human Factors in Computing Systems. Furthermore, we anticipate on publishing >25 peer-reviewed papers in scientific journals, as well as journals targeted at specific professions. Knowledge will also be exchanged with other ongoing research activities of the consortium, including Dutch Cardiovascular Alliance, IMPRESS consortium, CardioLab, 4TU project Pride & Prejudice (chronic disease prevention), Healthy PregnancyForAll, Loyalty program, eHealth programs as www.SmarterPregnancy.co.uk, ACE Pregnancy and childhood, Medical Delta Institute of maternal, fetal and neonatal medicine, ErasmusMC Koers23 and its Future Proof Strategy.

Students and early career researchers

BSc/BA/MSc/MA students will be **engaged** from the start. They will work on small projects, for which the appointed PhD-students and postdocs will formulate research questions appropriate for bachelor and master theses. Supervision will be done by PhD-students, postdocs and more senior-level researchers. We have already in place a ‘Proposal Format’ to organize this type of student engagement. Finally, at least 3 postdocs and 8 PIs from the already running Convergence Digital Twin project are involved in HELIOS. Within the ErasmusMC Graduate School for MSc/PhD students

and postgraduates, we will organize modules for Health&Tech lifestyle care using a life course approach.

Our WP leads are talented, almost all early/midcareer researchers, while within the WPs there is a mix of junior and senior researchers/professors. The early career researchers will be coached by our senior researchers and professors, paving the way to their engagement and help prepare them for their future scientific careers. This will have a significant impact on their future chances of tenure track.

Attracting talents

We will adhere to the advice of the ErasmusMC Standard Evaluation Protocol and Strategic Talent Management Plan and the general application procedures for attracting young talents at EUR and TUD. This means that we will actively recruit scientific diversity in terms of gender, demographics and academic profiles, and will attract talents from within, as well as outside, the current participating organizations.

During the project, we will ask the talent classes (e.g., Female Career Program) and coordinators of these classes at ErasmusMC ('Talent and Innovation Council'), EUR and TUD to inform potential recruits about our project.

Attracting funding

We will create and expand our eco-system with specifically societal stakeholders, and deliver a blueprint for the future development of other LCT. This will provide a strong position for new grant-applications, partnerships with **external financiers**, and the generation of start-ups.

2.e Long term societal impact

In the Netherlands, 1.5 million people suffer from CVD resulting in a loss of health and QoL. When we describe this as Disability-Adjusted-Life-Years, CVD counts for 16% of the total direct expenses in healthcare. The prognosis for the coming decade is that the number of patients with CVD will increase, due to increasing life-expectancy, better survival after CVD events, the obesity epidemic, and the consequences of COVID-19. HELIOS will tackle this global health challenge by optimizing the 'healthy ageing process' through **four pathways** leading to our expected **impact**. The Impact Pathway is described in detail in Figure2, presenting the **output** delivered by HELIOS, the uptake of the results by relevant stakeholders resulting in **outcome**, and long-term **impact** for **healthcare, society, science and economy**.

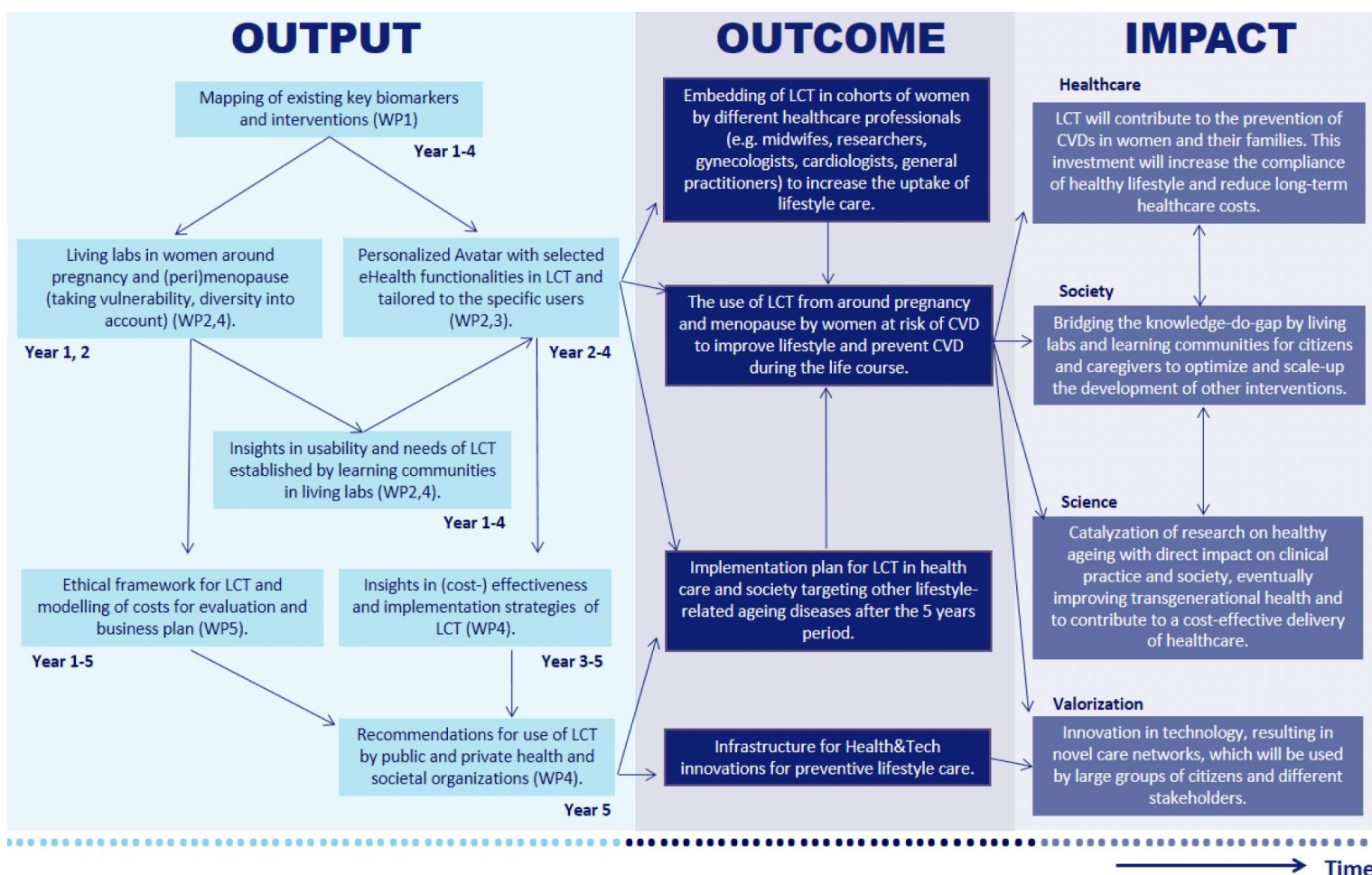


Figure2. Impact pathway

In the long term, LCT will benefit society by integrating hospital care with care at home and in the living environment, saving both patients and professionals time and resulting in a higher quality and efficiency of care. We anticipate on an increased 'sense of urgency' among professionals and policy makers on a municipality and national level regarding the importance of providing combined genetic, lifestyle and environmental pathways. The enhancement of technology by the development of LCT will create innovative and unusual new partnerships and novel care networks, within our own hospital, but also for other (inter)national hospitals.

To **measure impact** we distinguish the following categories: a) societal relevance, b) collaboration between universities and disciplines, c) publishing in peer-review journals and highest quartiles, d) adherence to Open Science.

Expected TRLs and necessary partners

The design of LCT will be iteratively tested and implemented in the HELIOS cohorts. Each WP covers different technology readiness level scales:

WP1 focuses on fundamental, translational and clinical research (**TRL 1-2**). This will be used to outline design requirements for the development of biomarkers that will feed into LCT. **WP2** Addresses the iterative design and evaluation process (**TRL 3-7**) that will use the integrated knowledge from WP1 to design targeted interventions that promote healthy lifestyles. The context for testing will move from evaluations in living labs to user studies in the operational environment (patient's home and hospital). The validation will focus mostly on aspects of user-acceptance.

WP3 focuses on data collection (e.g., medical records, sensors, wearables), development and validation of prediction models and risk scores, and a viable software architecture for this (**TRL 3-7**). This will form the base for patient-Avatar interaction and provide inputs for behavior change interventions to achieve healthier lifestyle. **WP4** implements, validates and evaluates each of the LCT prototypes (**TRL 7-8**), targeting a version deployed and implemented in the Digital Connected platform of **WP2 (TRL9)**. **WP5** focuses on ethical alignment of data collection, algorithmic decision making, prototyping, and how LCT influence users for which we will apply meaningful human control principles and principles of legitimate interpersonal influence. This WP will also demonstrate the system's economic feasibility in terms of (cost-)effectiveness (**TRL 3-9**).

3. Flagship proposal (25%) max 2000 words for 3.a to 3.d (total word count 1985)

3.a Ambitions

HELIOS proposes a unique research strategy combining fundamental, translational and behavior research on healthy ageing and integrating results into implementable solutions accepted by a wide range of stakeholders. Our ambition is supporting the strategic themes for a 'future-proof' ErasmusMC: Equity of care, Lifestyle, Ageing and Societal impact. Also, the ambitions of the ErasmusMC Sophia Children's Hospital for a life course approach of care and centralization of CVD care in the 'ErasmusMC Cardiovascular Institution' fits perfectly with HELIOS.

Essential stakeholders from our Convergence Digital Twin project are on-board in HELIOS, distinguishing our program from other (inter)national research consortia and institutes. Current ageing institutes focus on fundamental biology of ageing. While fundamental research often fails to transmit its findings to human-centered ageing research, the latter often fails to implement findings within society at large due to the lack of a personalized information ecosystem representing all relevant stakeholders. To generate maximum impact, an all-inclusive knowledge chain is required to translate and implement scientific and technological advances from bench to bedside and into society. We will create this essential ecosystem for LCT as roadmap for life course journeys, with medical, technological, lifestyle, environmental and societal domains integrated. This will tackle the greatest challenge, namely realizing sustainable adoption of digital and blended lifestyle care combined with medical and social care in healthcare and society, in particular vulnerable populations. Altogether, HELIOS is perfectly suited to achieve the Convergence Health&Tech ambitions of healthier lifespan, reduced health inequalities and affordable healthcare system.

3.b Aims and feasibility

HELIOS **projects' aims** are:

- WP1. Identification of key biomarkers and interventions for CVD.
- WP2. Assessment of needs/values of target women, families and caregivers for content of LCT on the Digital Connected platform.
- WP3. Design of AI-based CVD-prediction models and manufacturing and testing of new lifestyle care solutions for LCT.
- WP4. Implementation of LCT for target women and evaluation on usability, effectiveness, and (inter)operability.
- WP5. Delivery of ethical framework for meaningful human control in the algorithmic generation of LCT and technology, and estimation of cost-effectiveness from a societal perspective.

Strategy to involve students in research plan, See 2.d and 3.c.

Feasibility

HELIOS builds on unique knowledge and infrastructure created in the 2 years Convergence project Digital Twin. The cohorts incorporated for living lab experiences (WP2) and implementation in healthcare and society (WP4) realistically represent our target population. Notably, the Digital Twin staff and part of its external co-funders join HELIOS, e.g., Microsoft-NL develops ErasmusMC Digital Connected platform and Microsoft-Europe additionally invests in LCT.

Risk mitigation measures

Poor end-user involvement. We will collaborate with a client panel, event-organization 'Improve', and municipality of Rotterdam in reaching the end-users. Living labs with learning communities in WP2 will facilitate this level of involvement.

Poor data quality. The success of applying AI depends on the quality of large data volumes. High-quality data volumes are ensured by existing large well-annotated data sets and other datasets to be added. Moreover, we have ample experience to make data interoperable, and connect data across data shields using personal-health-train-solutions.

Unfavorable usability profile. The co-design of LCT, as complex lifestyle care intervention with end-users in living labs and learning communities, will ensure that LCT will meet their needs and values. LCT building blocks will be selected based on evidence and proven quality, guaranteeing an effective intervention.

Attracting external funding

Considering the track records of HELIOS consortium, current co-financing >Euro 3.400.000 and potential co-financing (Annex II), we are highly capable to acquire funding and to set-up public-private-research partnerships. The current interest by a vast amount of private and societal partners will greatly boost future grant-proposals and result in further financial contributions by industries.

Our roadmap for a long-term sustainable funding level also foresees in the continuous mapping of internal and (inter)national funding opportunities and proposal submission. WP-leads and post-docs will be in the lead of the funding roadmap. Postdocs will prepare grant-proposals, supported by senior researchers and PhD, master and bachelor students.

3.c Work plan

See also Figure3.

WP0. Coordination and program management

Administrative, digital, financial, and legal management. Monitoring of dissemination and exploitation.

WP1. Fundamental and translational research on biomarkers and interventions

1. Mapping of biomarkers and physical characteristics in HELIOS intervention cohorts.
2. Define combinations of key-ageing markers and preventive measures from multimodal data sources from preconception until postmenopausal age as Big Data for WP2&3.

Non-exhaustive list of candidates: bloodpressure, weight, activity/exercise, rhythm, menstrual cycle, cardiovascular history, nutrition/lifestyle, chronological/biological age, (stress)hormones, (pharmaco)genetic, metabolomic, immunology, microbiome, microplastics.

<i>Deliverables & Milestones:</i>	Y1	Y2	Y3	Y4	Y5
D1 Map of existing biomarkers and physical characteristics.	X	X	X		
M1Big Data transfer to WP2&3.				X	
<i>Student projects (BSc/BA/MSc/MA)</i>	1	2	2	1	
<i>PhD 1.0FTE</i>	X	X	X	X	
<i>Postdoc 0.5FTE</i>	X	X	X		

WP2. Co-design and evaluation of LCT and integrated Avatar-based care interventions

1. Demonstrate how technology-enabled applications can support women and clinicians in LCT.
2. Demonstrate how technology-enabled interventions can adapt to diverse interacting users, resulting in better representation, increase ease of use, and more (long-term) engagement.

LCT building blocks are a) apps, wearables and tele-monitoring devices as interventions input, and b) interactive consultation methods, avatars and loyalty programs to deliver technology-enabled interventions. Co-design methods will be used in learning communities and living lab context. An Avatar service concept will be refined and validated through three iterative cycles. User-acceptability, usability, experience, desirability, and clinical appropriateness will be evaluated. Digital twin versus healthcare smart-assistant as Avatar variations will be assessed for appropriate representations of gender, ethnicity, and ableness. WP1&3 will feed into this WP to define the final LCT-concept for implementation on ErasmusMC Digital Connected platform and for evaluation (WP4).

<i>Deliverables & Milestones:</i>	Y1	Y2	Y3	Y4	Y5
D2 LCT Blueprint.		X			
D3 Recommendations to nudge behavior change in LCT interventions.	X				
D4 Three Avatar-based app prototypes with increasing fidelity level.		X	X	X	
M2 Validated product-concept for Digital Connected platform and WP4.			X	X	
<i>Student projects (BSc/BA/MSc/MA)</i>	1	2	1	2	
<i>PhD 1.0FTE</i>	X	X	X	X	
<i>Postdoc 0.5FTE</i>	X	X	X		

WP3. Secure integration of mixed data and designing interpretable risk prediction models

1. Enable LCT manufacturing by generating comprehensive longitudinal data prediction models based on existing (WP1) and prospective (WP2&4) data.

2. Generalize prediction models to individuals and offer clinical decision support through risk scores.

Novel statistical/machine learning approaches will be applied that can deal with heterogeneity, non-overlapping/paired data, sparseness and class-imbalances of data trajectories. To ensure appropriate responsibility of women, professionals, and designers over advice and actions of the automated AI-systems, a meaningful human control over prediction models is designed. To ensure interoperability, data will be mapped onto a common data model (FAIR, OMOP, HL7v2, FHIR). For data sharing, synthetic data schemes (adding noise) and federated learning schemes (Health-RI) will be employed. Prediction models will be applied to CVD and iteratively updated based on how individualized predictions help women in a test-environment.

<i>Deliverables & Milestones:</i>	<i>Y1</i>	<i>Y2</i>	<i>Y3</i>	<i>Y4</i>	<i>Y5</i>
D5 Fairification schemes of LCT data and privacy analyses procedures.	X	X			
D6 Algorithms to integrate and analyze heterogeneous data.	X	X	X		
D7 Prediction models in women around pregnancy and menopause for cardiovascular health score and value-based outcomes.			X	X	
M3 Refined prediction models based on first LCT applications.				X	
<i>Student projects (BSc/BA/MSc/MA)</i>	2	2	2	1	
<i>Postdoc 0.8FTE</i>	X	X	X	X	

WP4. Evaluation and implementation of LCT

1. Test and implement LCT as lifestyle intervention in HELIOS cohorts.

2. Evaluate LCT for feasibility, usability and outcomes.

Effectiveness will be evaluated with the Cardiovascular Health Score (CHS) and Patient Reported Outcome Measures (ICHOM).

The women of the cohorts* are invited for the learning communities/living labs (WP2) and evaluation and implementation of LCT in 'real life' in the HELIOS intervention cohorts embedded in the infrastructure of the same cohorts*.

*Rotterdam Periconception Cohort (250 HELIOS inclusions/year; pregnant women), Generation-R Study (1000 HELIOS inclusions/year; menopausal women), Generation-R-Next Study (1500 HELIOS inclusions/2years; pregnant women), FollowUpPreeclampsia (150 HELIOS inclusions/year; menopausal women). Considering the current follow-up of >15 years of the pregnant women of these cohorts, we expect 10-15% CVD: recruitment of 1000/year with 1 year intervention period in HELIOS intervention cohort.

Primary outcome is the change in American Heart Association's CHS. In our pilot-study of 244 women with preeclampsia, 1 year after pregnancy, the mean CHS was 10.0, with a standard deviation (SD) of 1.8. Thus, enrollment 1000 women, meaning before-after differences ≥ 0.1 SD (0.18 points) can be identified with 90% power (two-sided test with $\alpha=0.05$). This sample size is sufficient to enable multivariable analyses of (potential) determinants of CHS (changes).

<i>Deliverables & Milestones:</i>	<i>Y1</i>	<i>Y2</i>	<i>Y3</i>	<i>Y4</i>	<i>Y5</i>
D8 HELIOS intervention (LCT) study in 'real life'.		X	X	X	X
D9 Insights LCT usability and feasibility.			X	X	X
D10 Primary and secondary outcomes after LCT intervention in end-users.				X	X
M4 Recommendations and roadmap for use of LCT in CVD and application to other diseases.				X	X

<i>Student projects (BSc/BA/MSc/MA)</i>	2	2	2	2
<i>PhD 1.0FTE</i>	X	X	X	X
<i>Research nurse 0.8FTE</i>	X	X	X	

WP5. Ethics and Health Policy & Costs

1. Assess LCT from an ethical perspective and evaluate cost-effectiveness from societal perspectives.
2. Evaluate 'output' of LCT, ensuring that technology-mediated influences are ethically sound, autonomy-preserving and non-manipulative.

To assess the impact of LCT on patient autonomy, well-being, meaningful human control and manipulation, we will use theoretical frameworks from moral philosophy and the philosophy of (AI) technology, and meaningful ethical control framework developed also at TUD. The latter deals with ethical risks associated with AI-based technologies, e.g., privacy loss, explanatory opacity, or responsibility gaps. Insights from WP1&3 will feed WP4.

Cost-effectiveness analysis will explicitly consider effects of LCT interventions, healthcare costs and broader societal outcomes as labor market participation.

<i>Deliverables & Milestones:</i>	Y1	Y2	Y3	Y4	Y5
D11 Ethical risk mapping for generating health recommendations on sensitive health and lifestyle data.		X			
D12 Guidelines and principles for meaningful LCT ethical framework.		X			
D13 Ethical principles for legitimate technology-mediated user influence in sensitive health-contexts.		X	X		
D14 Insights in relation between LCT and healthcare consumption/labor market participation/informal care.			X	X	X
D15 Model based cost-effectiveness analysis of LCT.				X	X
M5 Strategic and cost-effective plan for LCT implementation.			X	X	X
<i>Student projects (BSc/BA/MSc/MA)</i>	1	1	1	1	1
<i>2 Postdoc 0.15FTE</i>	X	X	X	X	X

Required facilities and infrastructure plan

HELIOS will use existing data from the listed cohorts for LCT development and collect prospective data on LCT validation in the same cohorts. Local data storage will adhere to institutional data management policies ensuring data availability during and after the project. Each WP will develop protocols for standardized data use, with WP3 specifically for data definition/code books, collection, harmonization, creation and evaluation as well as reusable syntax files. WP leads hold primary responsibility for data collected in their WP. We will agree on specific protocols for informed consent and data privacy, and a system for participants to report unethical behavior or problems. The three Flagship leads will govern the source data used and all necessary data collection methods.

3.d Alignment of the program with relevant national and international research agendas

This is a high-risk high gain project leading to novel insights in CVD prevention in women, from the start of life onwards and across the life course. HELIOS fully aligns with the Dutch National Research Agenda and the Netherlands Federation of University Medical Centers agenda, with specific objectives for use of eHealth in healthcare and sustainable healthcare at the highest level.

Internationally, our project contributes to the UNs Sustainable Development Goals, specifically regarding reducing gender inequalities through our focus on women's health across the life course. Our strong direction on the development and integration of multiple technological tools into novel LCT, serving as roadmaps for life course care journeys in future health care, aligns with local, national and European strategies to integrate eHealth strategies in future health care. While HELIOS has the biggest contribution in life sciences, it also aligns with research in marketing and business on theory development for use and effectiveness of Avatars to nudge behavioural changes as well as on human-machine interaction and how consumers trust and respond to being serviced by machines versus humans.

4 Consortium (25%) max 2000 words for 4.a to 4.d (total word count 1894)

4.a Track record of the consortium members

Our consortium has an outstanding track record regarding research, translation of results into applications, supervision of students and funding acquisition, as shown in the overview below for the main PIs.

Applicant	Field of Expertise/relation to the aims	Translation of scientific results	Publication number
			H-Index
			BSc /MSc supervision
			External funding amount (€)
			Awards (N)
Steegers-Theunissen	Periconception, pregnancy, lifestyle-care, metabolism, epigenetics, epidemiology, PI Rotterdam Periconception cohort	Development and implementation of blended lifestyle care and digital lifestyle care innovations: 4 lifestyle coaching programs and 1 loyalty program on the mobile phone. www.SlimmerZwanger.nl / https://www.mijnleefstijlzorg.nl/	>350
			58
			>120 MSc, 35 PhD
			>20M
			21
Van Everdingen	Marketing and business applications of new technologies across the customer journey. Offering insights how to optimally design a life course care trajectory.	Marketing and implementation of new, high-tech products. Knowledge about how co-creation and gamification can help the development of new products to increase acceptance.	30
			14
			>200
			50K
			1
Abbink			>160

	Human-Automation interaction and Human Factors. To ensure user engagement and responsible use of technology for diverse users with a contribution to meaningful human control over AI-based systems.	Three times tech-transfer to market: 1) developed world's first haptic gas pedal marketed by Nissan; 2) contributed to worlds 1st steer-by-wire car; 3) contributed to lane-keeping assist systems. Multiple spinoff companies from PhD and MSc students.	33
			104 MSc, 9 PhD
			9.7M
			19
Hays	Microbiology, infectious disease detection and antimicrobial resistance. Access to bacterial, mycological and parasitological infection diagnostics, research facilities and protocols.	Contributed to AMR Awareness day and humans of Global Health in 2016.	110
			22
			<10
			12.5M
			-
Roeters van Lennep	Women's health, CVD and lipids.	Contributed to women's health and international CVD guidelines, founded clinic for cardiovascular follow-up after preeclampsia.	148
			28
			>25
			2.9M
			5
Luce Lupetti	Design Research, Participatory technology development.	Contextual and co-creation research for developing usable and acceptable applications of advanced technologies (robotics for cultural heritage, education).	38
			7
			>10
			5.8K
			2
Reinders	Expert in applying cutting-edge machine learning to develop data-driven analysis methodologies with applications in the health domain.	Introduced algorithm (WISECONDOR), that can detect chromosomal aberrations in fetal DNA present in maternal plasma from low-coverage whole genome sequencing data. This is being used now in routine clinical testing (NIPT) in hospitals in the Netherlands as well as abroad (e.g., France, Korea).	602
			66
			>200
			>10M
			-
Rousian	Pregnancy, lifestyle, embryo, epidemiology.	Impact of lifestyle factors on embryonic/fetal growth and development; construction of	36
			14
			>10

		imaging atlas of embryonic brain development; implementation of innovative virtual reality imaging in pregnancy into clinical practice.	650K
			4
Gaillard	Obesity, CVD and developmental programming.	Development of app with reference charts for gestational weight gain for clinicians and researchers.	138
			32
			>10
			>1M
			1
Klenk	Ethics and philosophy of technology. The ethical aspects surrounding data collection and analysis as well as permissible and impermissible ways to influence patients (e.g., through apps).	Developed and implemented ethical software default settings for the leading open-source developer of Conversational AI.	44
			8
			<10
			610K
			5

Furthermore, a summary of the track records of the applicants per WP is presented. **WP1 applicants** have a total publication number >2500, together supervised >350 BSc/MSc students and have a total external funding amount >50M. Examples of translation of scientific results of relevance for the HELIOS flagship are: the international standardizations of diagnostic immunological assays and successful marketing of immunogenetic assays, blood-biomarker based prediction models for individualized prognostication in heart failure patients. **WP2 applicants** have a total publication number >150, supervised >65 BSc/MSc students and have a total amount regarding external funding of >4M. Examples here are development of technology-based solutions for biomedical and consumer applicants, applied co-design to develop various health services for CVD patients. **WP3 applicants** have a publication number >1200, supervised together >150 BSc/MSc students and have a total external funding amount >50M. Examples regarding translation of scientific results are building digital twin models for diagnosis and prognosis using AI, development of an ECG analyses that is marketed, and development of open source solutions for predictive analytics. **WP4 applicants** have a total publication number >900, supervised >50 BSc/MSc students, and have a total amount of >10M regarding external funding. Examples are clinical implications regarding the cost-effectiveness of the use and implementation of several medical screening tests. Finally, **WP5 applicants** have a total publication number >200, supervised >60 BSc/MSc students and have a total amount of >2.7M. An example is changing guidelines for economic evaluations in healthcare and recommendation of inclusion of future medical costs.

4.b Expertise of the consortium members in relation to the aims and objectives of the program

Fit between aims and expertise HELIOS consortium

Our consortium consists of quadruple helix partners, namely: female citizens, researchers, doctors and societal organizations/partners from the public and private sector. Together we have a mission to enhance health and social equalities through fostering gene-lifestyle/nutrition interactions by making scientifically validated, accessible, user-friendly LCT that deliver personalized preventive

interventions for lifestyle-related adverse reproductive outcomes and ageing diseases (DOHaD) widely available for women.

All disciplines and expertise required to achieve this mission are represented in our consortium, namely 1. Academic and applied experts on (ageing)health, lifestyle, nutrition, genetics, epidemiology, social sciences, environment, human and bacterial genetics, immunology, digital and blended care interventions, AI, imaging, psychology, strategic design approaches for digital solutions, human-machine interactions, applied data analytics, meaningful human control, health economics, and ethics; 2. Designers of platforms, apps, gamification and Avatars; 3. End-users of women and healthcare and social care professionals; and 4. Private partners from industry required for societal and technological implementation.

Complementarity of members

Since we aim to target the ageing process in women from different perspectives and fields, a wide range of experts are present within our consortium. When the main Flagship leads (RST, YvE, DA) together with co-lead (MR) started with designing study aims and targets, it was clear that we needed several complementary members to fulfil these aims. In our organization we divided our experts within teams with one (or two) WP leader(s) and an advisory board, where overall the main leads will have the function to support a flexible and circular workflow to anticipate on the complexity of our large-scaled, interdisciplinary, multi-stakeholder project. The workflow will be directed into a coherent co-design process characterized by commitments to a collective prototype.

Commitment with other flagships/consortia

Important commitments, also with regards to external partners, and other flagships are present. These are: 'Human-centered AI for sustainable critical care' (Meeteren), 'Personalized genetic therapies for rare diseases' (Uitterlinden, Schaik), 'Continuity of care from hospital to home' (Ahaus), 'Deep medical imaging of structure, physiology and function' (Niessen), 'Tailoring technology for healthy neighbourhoods' (Steegers-Theunissen), 'Immunomics&eHealth' (Reinders).

4.c Involvement of partners

Involved partners and stage of involvement

Co-funders: Commitments for collaboration have been received from the following partners:

Municipality of Rotterdam, Rotterdam University of Applied Sciences, who are running many projects to increase health, QoL and improve living environments, e.g., KIC SmartFamilyBuddy, Healthy pregnancy for all, and (public) partners with expertise on lifestyle, health, eHealth, environment, interventions, dissemination and implementation.

Our societal partners from public and private sectors will have a role in the access to eHealth interventions, stakeholders and end-users. They can advise regarding their expertise with the end-users and in dissemination and implementation strategies (**TNO, RIVM, Medical Delta, Trimbos, Health Holland, Voedingscentrum**). IT companies (**Microsoft Europe/NL, Avegen Health, Soulive innovations**) will help to develop the technical functionalities of LCT, including Apps and, Avatars. Technology companies as **Corsano health, Merck Group and Somnox** will provide technological tools, wearables for research and monitoring. Health insurance companies and ErasmusMC department of Sales (**VGZ**) and **our institutional Technology Transfer Offices** will contribute to the development of a sound business and implementation plan.

End-users: Critical consortium members are women who will use LCT. They are already involved in this proposal via the **ErasmusMC Clientenraad and HELPP Foundation**. They will actively participate in the learning communities, co-creation of LCT building blocks and its first implementation.

Connection team members and external partners

All partners listed are involved in HELIOS already since the grant writing stage. There is a close interaction and exchange of ideas ongoing. This connection will be continued during the project, as warranted by our project governance structure. External partners will have a seat in the User committee, which will be invited for HELIOS consortium meetings and will be asked to give advice on the project directions and progress. Figure3 shows the connection within and between the WPs of team members and the external partners. The funding provided by the different external partners is also indicated per WP (see Annex II).

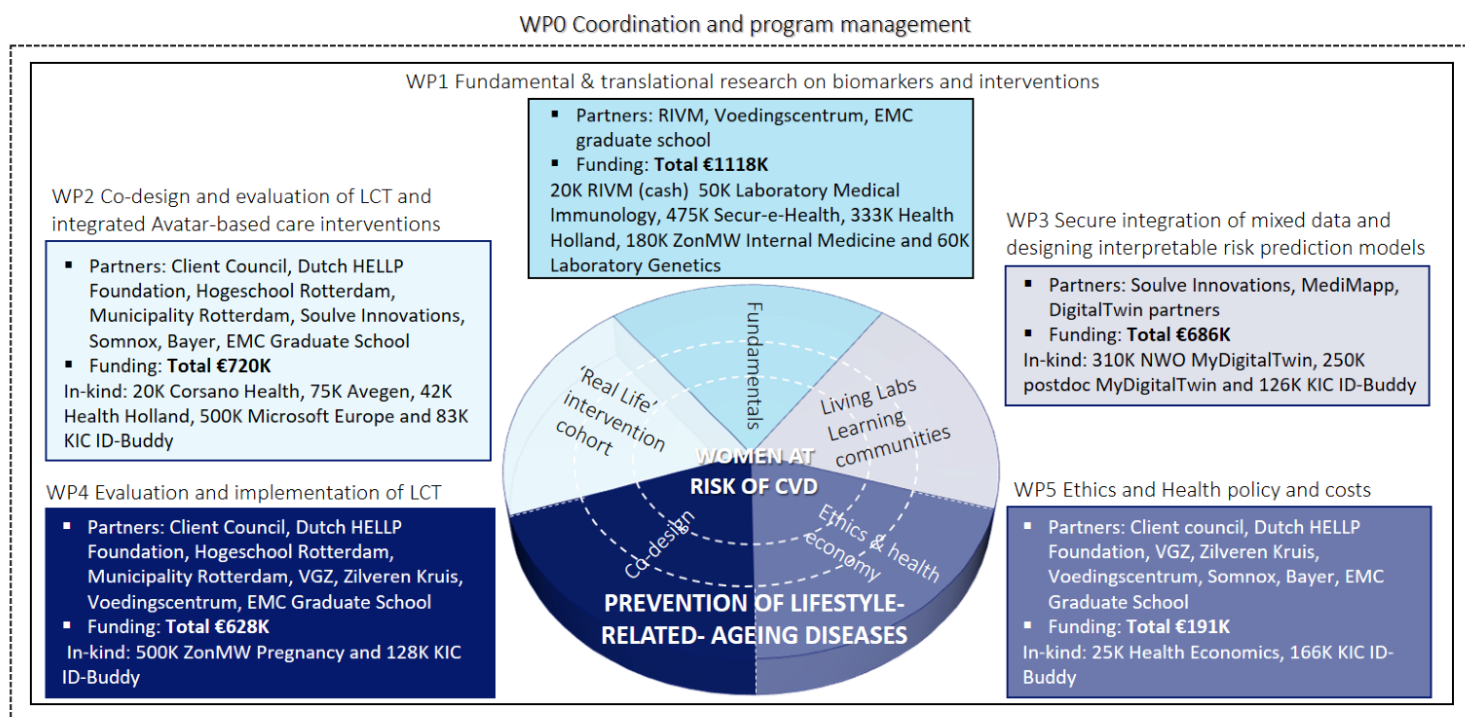


Figure3. Workplan of the development of LCT

4.d Governance, organization and collaboration strategy of the consortium

HELIOS will work in teams of WPs, with 1-2 WP leads from ErasmusMC, EUR and TU Delft. We have a WPO for coordination and program management, involving a program manager for the organization of this large-scale consortium. The program manager will work intensively with the HELIOS main leads. WP leaders are responsible for the progress within their WPs and will arrange for face-to-face or videoconferencing WP meetings at a frequency deemed necessary.

HELIOS is governed as follows:

Networking and project monitoring body: involving all consortium partners, meets 2x/year face-to-face to present results and discuss progress. This body is chaired by the Flagship main lead. Presentations by the PhD students and post-docs will be strongly encouraged.

Steering Team: involving WP leads and project manager, meets monthly face-to-face or by videoconferencing to monitor WP progress and solve hurdles and barriers. Issues for decision-making are part of this team. This body is chaired by the Flagship main lead.

Coordination Team: involving 3 main Flagship leads and program manager, responsible for day-to-day management, meets bi-weekly face-to-face or by videoconferencing, issues will be forwarded to the Steering Team.

User Committee: External partners, meets 1x/year with the Steering Team to monitor and steer the project together, is invited to join the meetings of the full consortium.

Advisory board: External experts on molecular genetics, pharmacogenetics, chronobiology, lifestyle, internal medicine, adapting new technology in clinical practice (**see Table 1. General Information**) to monitor project progress and advise the consortium. The board is invited to join the meetings of the full consortium.

Minutes of meetings will be shared with attendants of the relevant governance body to ensure transparency.

Annexes (do not count toward maximum number of pages)

I Budget table (including the external funding table)

See enclosed budget form.

II Letters of intent and letters of commitment

See enclosed the following Letters:

A Letters of intent

B Letters of commitment

C Letters of intent for future grant proposals

III Letter of support from department head of the main lead

See enclosed letter of the Head of the department of the main applicant.

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